

Engineering Chemistry-1

CHE109

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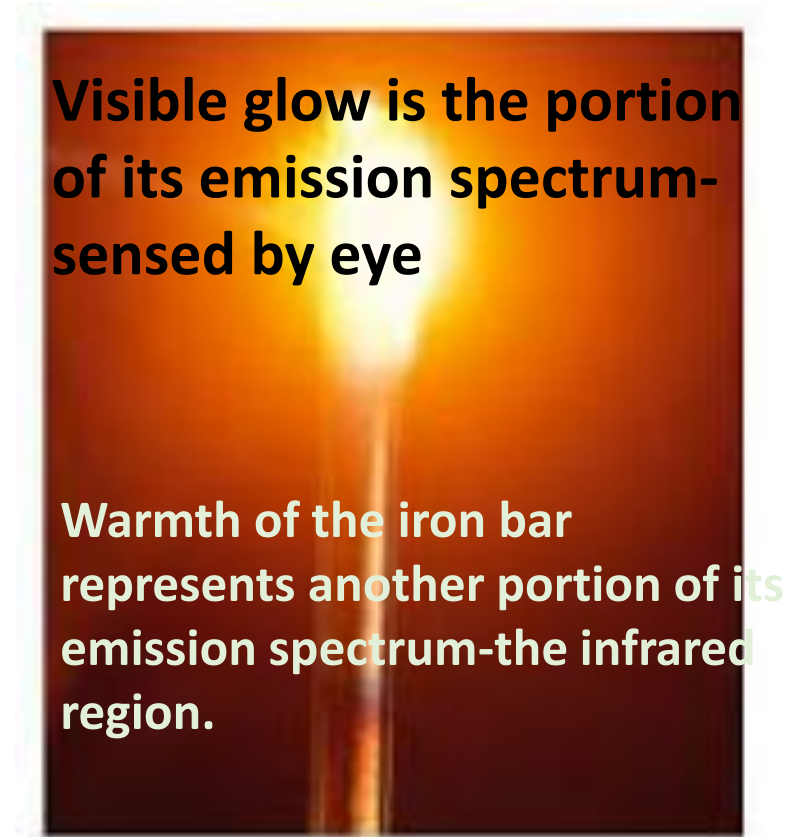
Emission Spectra

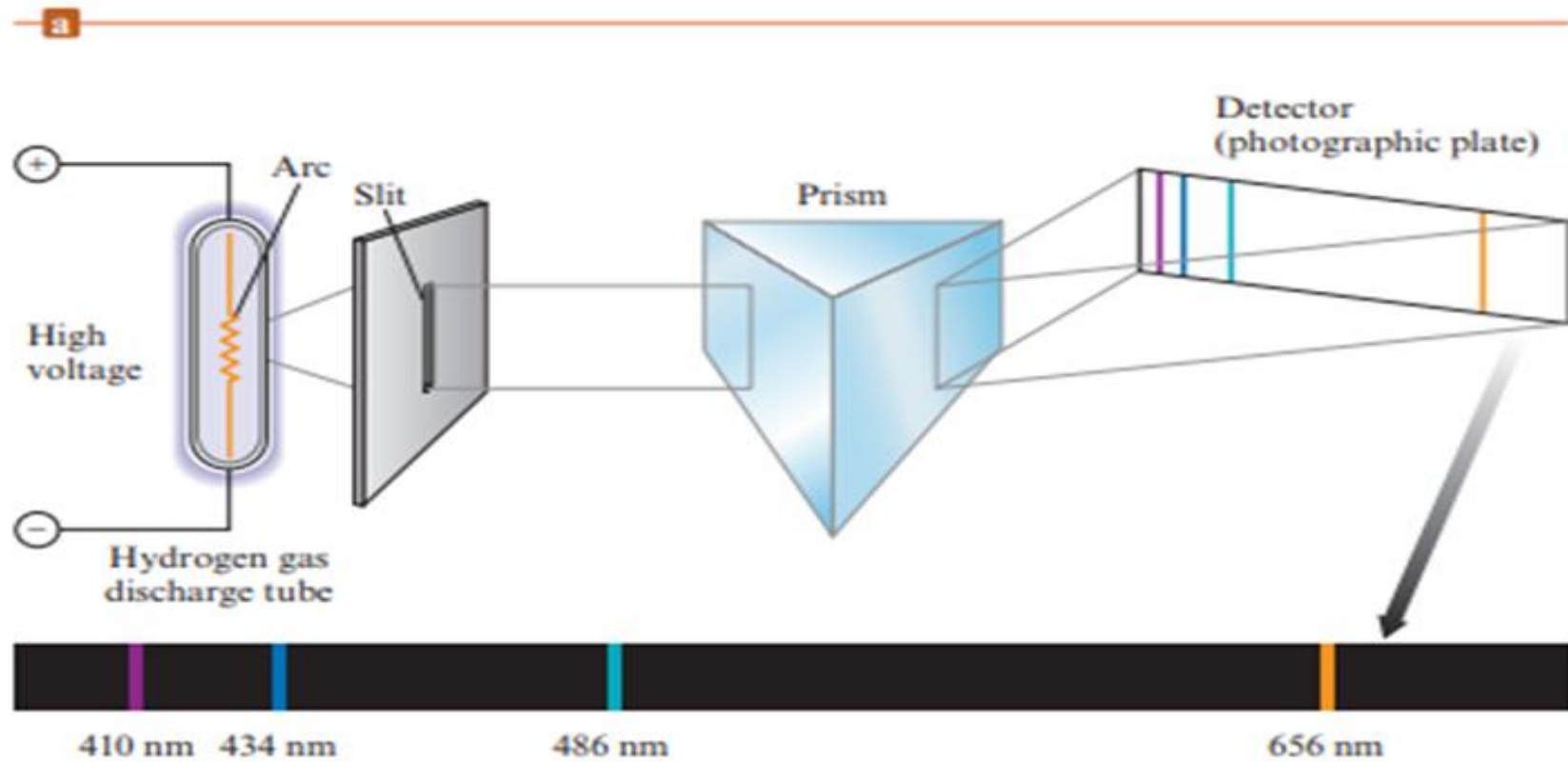
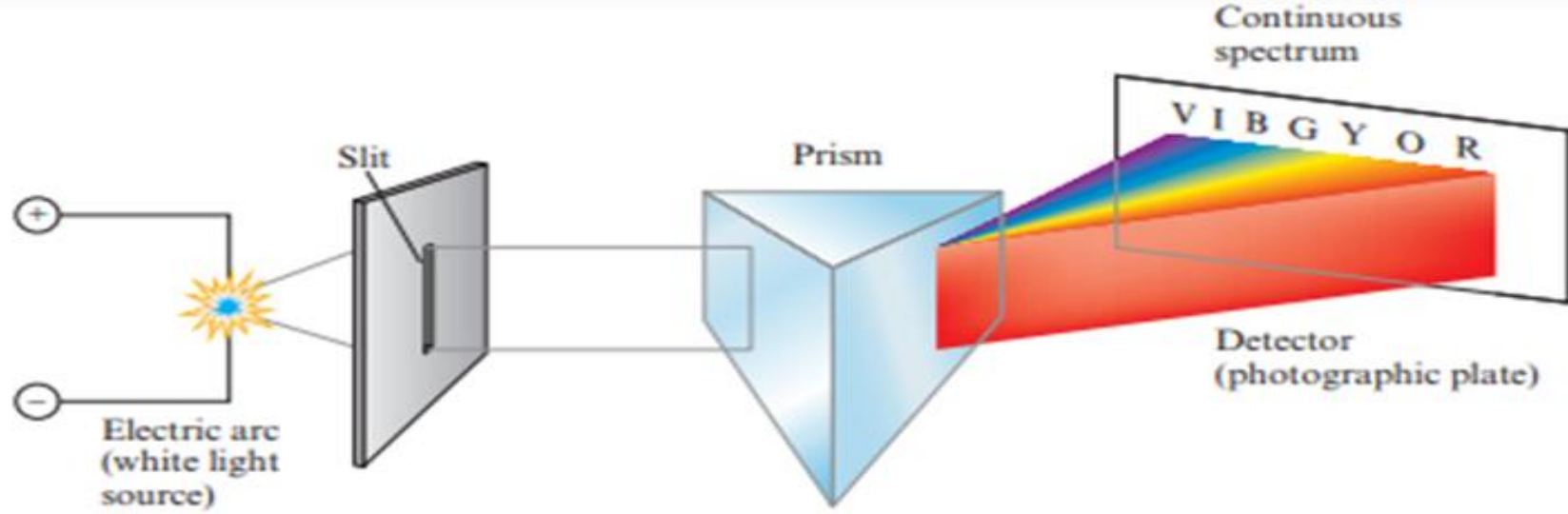
Study started ever since the seventeenth century about the emission spectra, that is, either continuous or line spectra of radiation emitted by substances.

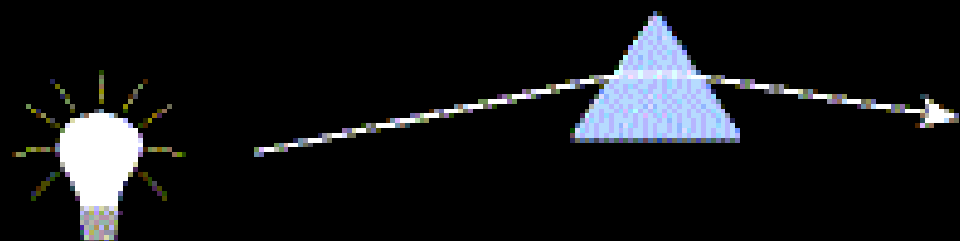
Energizing a sample of material either with thermal energy or with some other form of energy (such as a high-voltage electrical discharge).

EX; “red-hot” iron bar

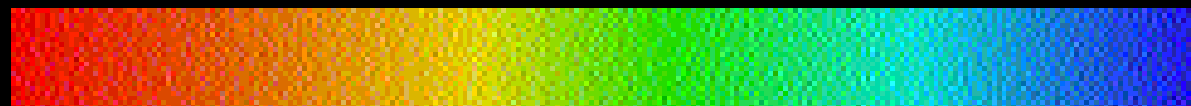
Portion of emission spectra (visible) of the sun and of a heated solid is that both are continuous;



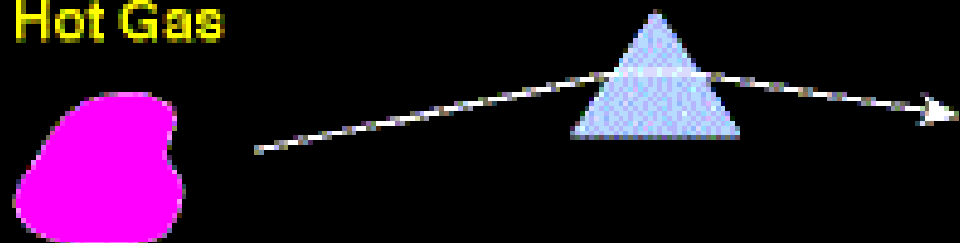




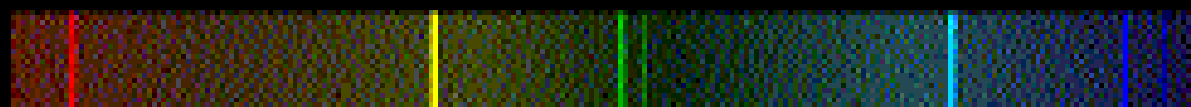
Continuous Spectrum



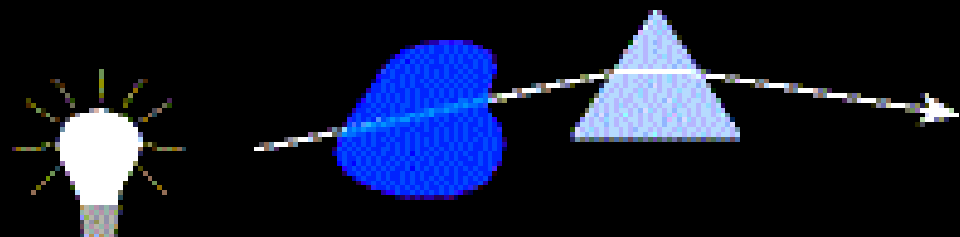
Hot Gas



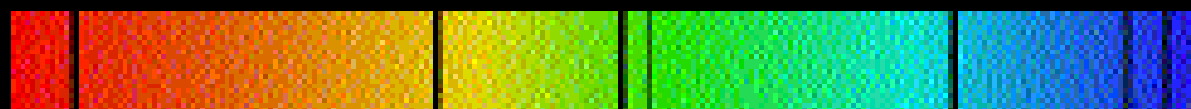
Emission Spectrum



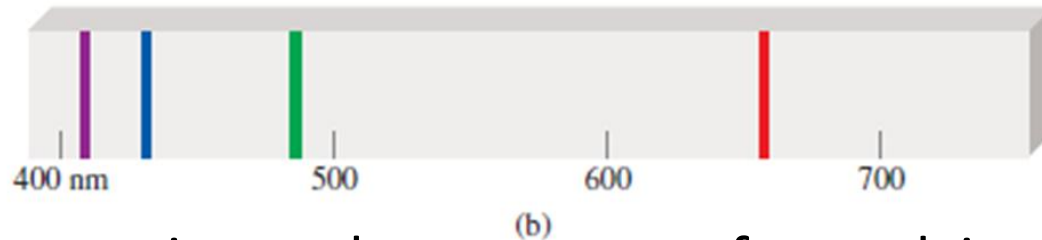
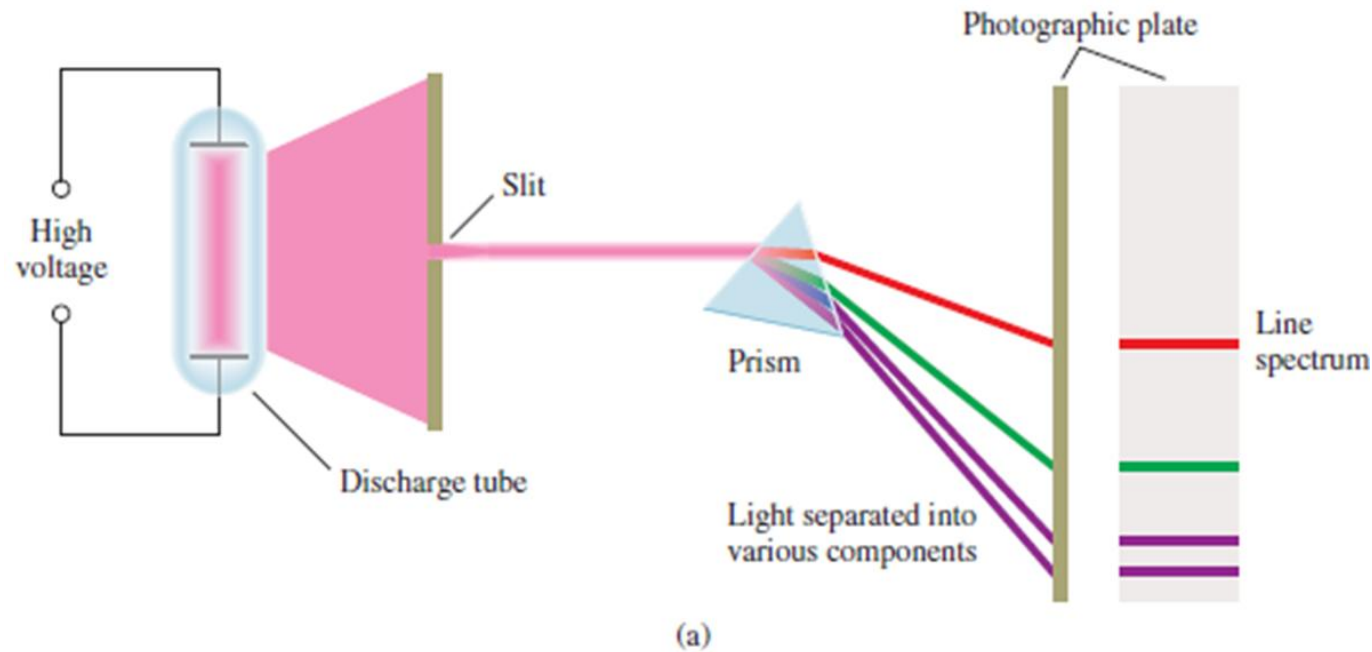
Cold Gas



Absorption Spectrum



Emission spectra of atoms in the gas



- (a) An experimental arrangement for studying the emission spectra of atoms and molecules.
- (b) The line emission spectrum of hydrogen atoms.

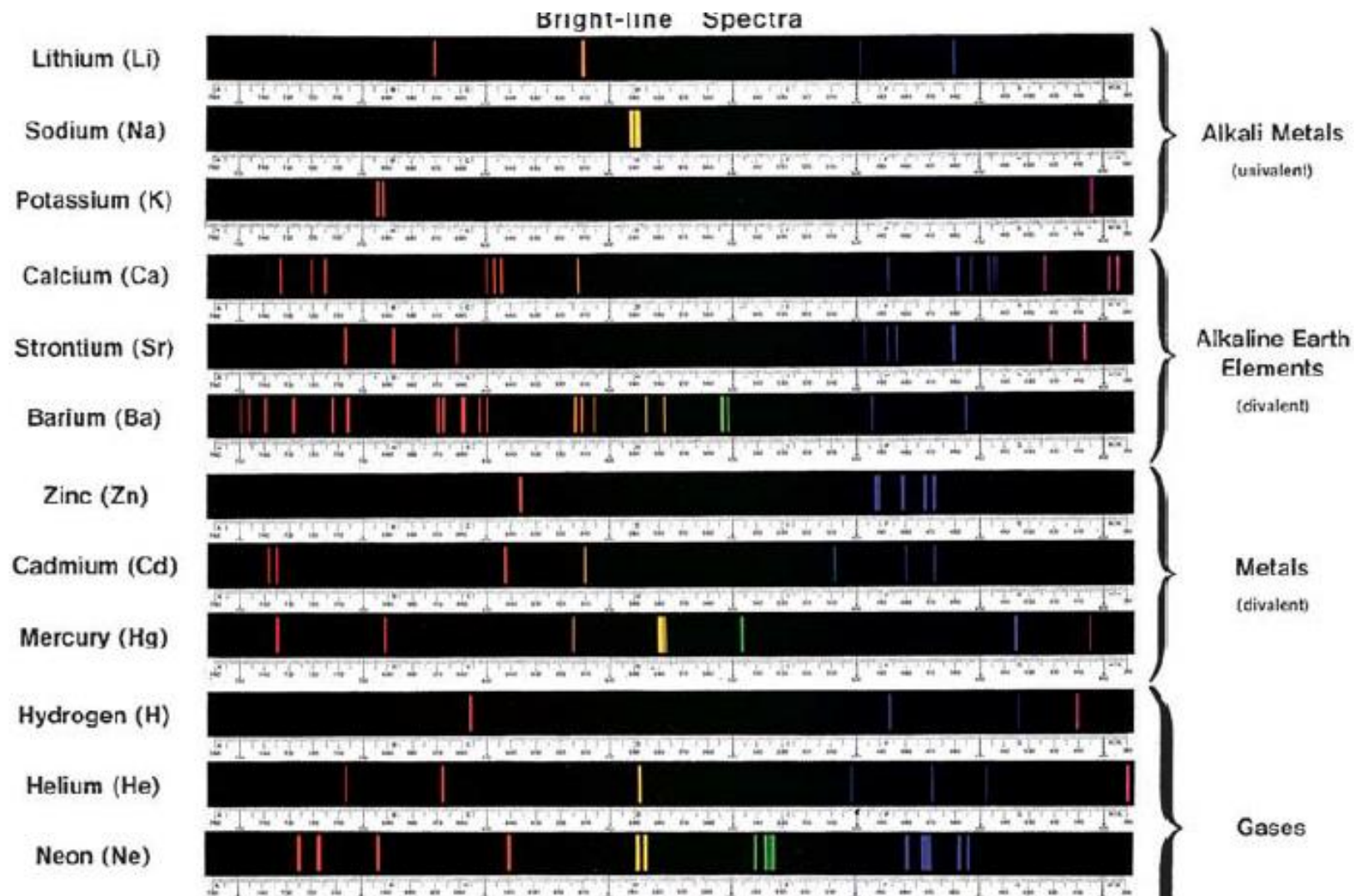
What about the emission spectra of atoms in the gas?

Not a continuous spread of wavelengths from red to violet;

Bright lines in different parts of the visible spectrum.

line spectra are the light emission only at specific wavelengths.

Every element has a unique emission spectrum. The characteristic lines in atomic spectra can be used in chemical analysis to identify unknown atoms, much as fingerprints are used to identify people.



Emission Spectrum of the Hydrogen Atom

What to consider:

- **Origination of emission spectrum of the hydrogen atom**
- **Weakness of Rutherford atomic model**

Bohr postulated that the electron is allowed to occupy only certain orbits of specific energies. In other words, the energies of the electron are quantized.



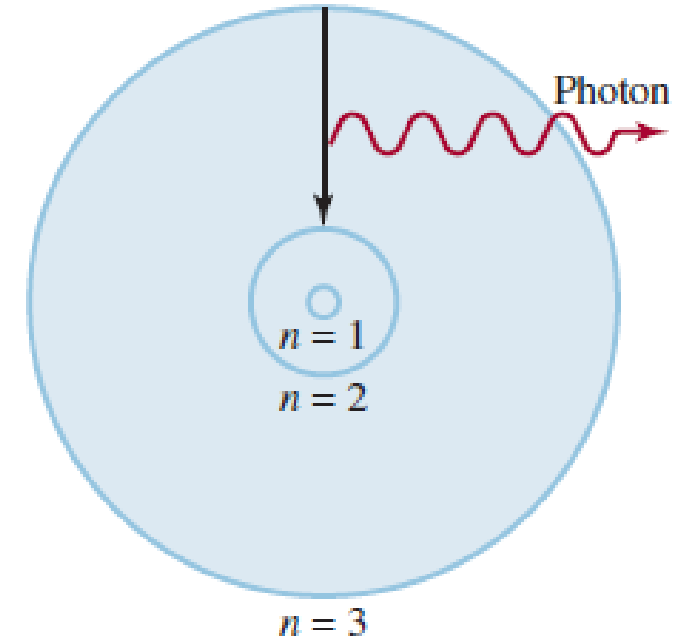
Bohr attributed the emission of radiation by an energized hydrogen atom to the electron dropping from a higher-energy allowed orbit to a lower one and emitting a quantum of energy (a photon) in the form of light

The energies that an electron in hydrogen atom can occupy are given by

$$E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$$

$$E_n = -R_H \left(\frac{1}{n^2} \right)$$

R_H , the Rydberg constant for the hydrogen atom, has the value $2.18 \times 10^{-18} \text{ J}$. The number n is an integer called the principal quantum number; it has the values $n = 1, 2, 3, \dots$



Why (– V_e) sign in the expression of energy

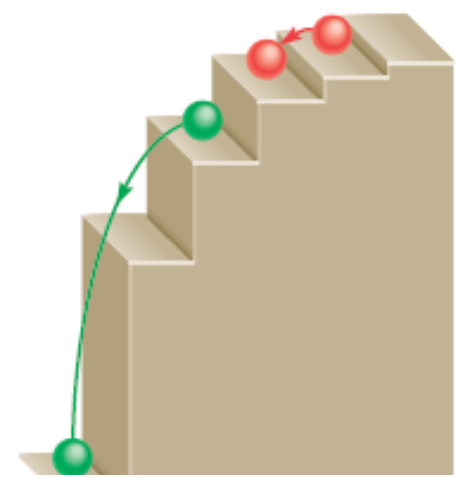
- Energy of the electron in the atom is lower than the energy of a free electron(infinitely far from the nucleus).
- Mathematically, this corresponds to setting n equal to infinity in Equation (page 7), so that $E_{\infty}=0$. As the electron gets closer to the nucleus (as n decreases), E_n becomes larger in absolute value, but also more negative.
- The most negative value, then, is reached when $n = 1$, which corresponds to the most stable energy state. (ground state, the lowest energy state) $n = 2, 3, \dots$
- Each of these levels is called an excited state, or excited level, which is higher in energy than the ground state.
- A hydrogen electron for which n is greater than 1 is said to be in an excited state. The radius of each circular orbit in Bohr's model depends on n^2 . nucleus).

Bohr's theory enables us to explain the line spectrum of the hydrogen atom.

Radiant energy absorbed by the atom causes the electron to move from a lower-energy state (characterized by a smaller n value) to a higher-energy state (characterized by a larger n value).

Conversely, radiant energy (in the form of a photon) is emitted when the electron moves from a higher-energy state to a lower-energy state.

The journey from a lower step to a higher one is an energy-requiring process, whereas movement from a higher step to a lower step is an energy-releasing process.



$$\Delta E = E_f - E_i$$

$$E_f = -R_H \left(\frac{1}{n_f^2} \right)$$

$$E_i = -R_H \left(\frac{1}{n_i^2} \right)$$

$$\Delta E = \left(\frac{-R_H}{n_f^2} \right) - \left(\frac{-R_H}{n_i^2} \right)$$

$$= R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

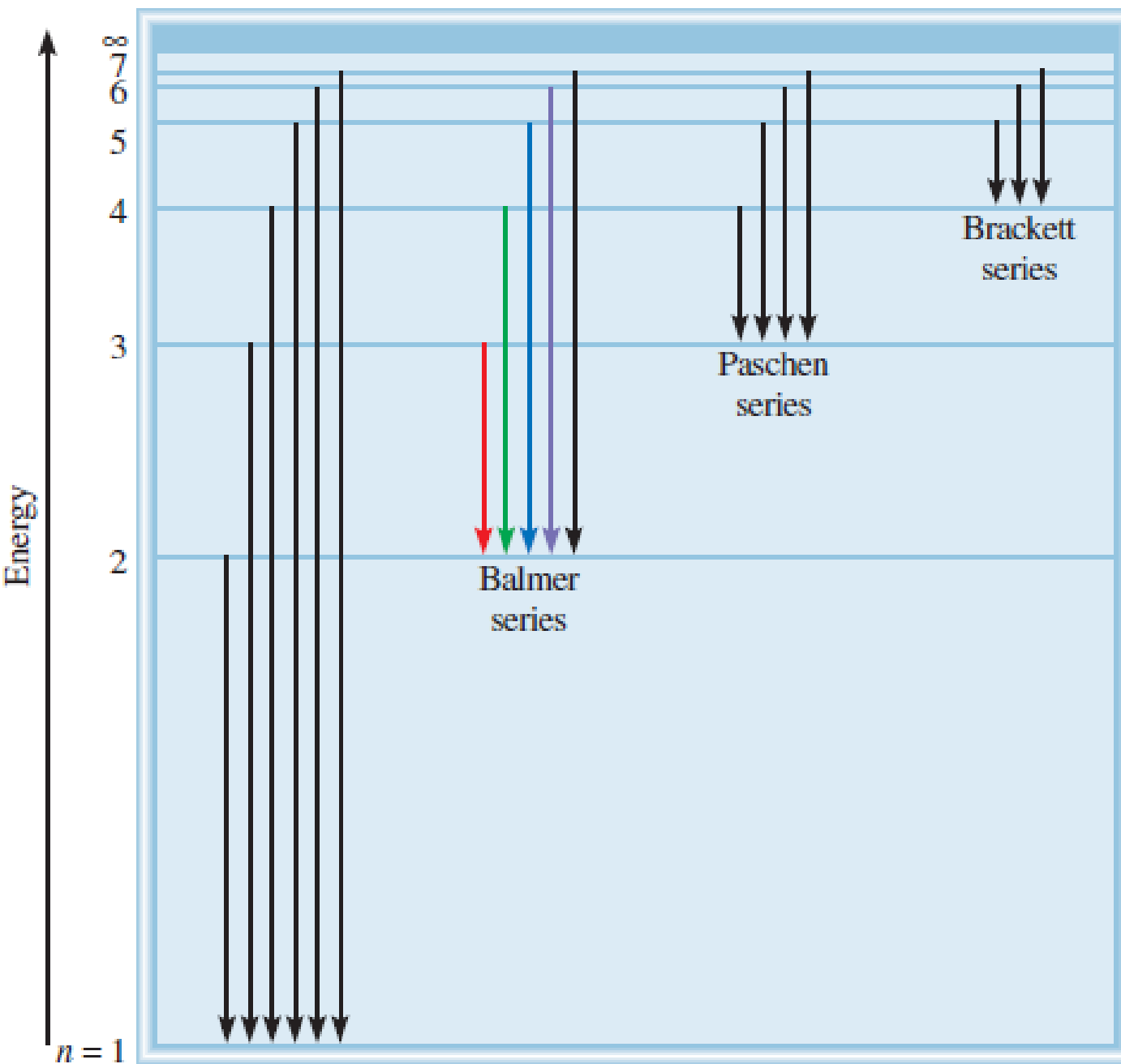


TABLE 7.1 The Various Series in Atomic Hydrogen Emission Spectrum

Series	n_f	n_i	Spectrum Region
Lyman	1	2, 3, 4, ...	Ultraviolet
Balmer	2	3, 4, 5, ...	Visible and ultraviolet
Paschen	3	4, 5, 6, ...	Infrared
Brackett	4	5, 6, 7, ...	Infrared

What is the wavelength of a photon (in nanometers) emitted during a transition from the $n_i = 5$ state to the $n_f = 2$ state in the hydrogen atom?

2627.66 nm

$$\begin{aligned}\Delta E &= R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \\ &= 2.18 \times 10^{-18} \text{ J} \left(\frac{1}{5^2} - \frac{1}{2^2} \right) \\ &= -4.58 \times 10^{-19} \text{ J}\end{aligned}$$

The negative sign indicates that this is energy associated with an emission process.

To calculate the wavelength, we will omit the minus sign for ΔE because the wavelength of the photon must be positive.

$$\begin{aligned}\lambda &= \frac{c}{\nu} \\ &= \frac{ch}{\Delta E} \\ &= \frac{(3.00 \times 10^8 \text{ m/s})(6.63 \times 10^{-34} \text{ J} \cdot \text{s})}{4.58 \times 10^{-19} \text{ J}} \\ &= 4.34 \times 10^{-7} \text{ m} \\ &= 4.34 \times 10^{-7} \text{ m} \times \left(\frac{1 \text{ nm}}{1 \times 10^{-9} \text{ m}} \right) = 434 \text{ nm}\end{aligned}$$

What is the wavelength (in nanometers) of a photon emitted during a transition from $n_i = 6$ state to the $n_f = 4$ state in the H atom?

Calculate the energy required to excite the hydrogen electron from level $n = 1$ to level $n = 2$. Also calculate the wavelength of light that must be absorbed by a hydrogen atom in its ground state to reach this excited state.*

Solution

Removing the electron from a hydrogen atom in its ground state corresponds to taking the electron from $n_{\text{initial}} = 1$ to $n_{\text{final}} = \infty$. Thus

$$\begin{aligned}\Delta E &= -2.178 \times 10^{-18} \text{ J} \left(\frac{1}{n_{\text{final}}^2} - \frac{1}{n_{\text{initial}}^2} \right) \\ &= -2.178 \times 10^{-18} \text{ J} \left(\frac{1}{\infty} - \frac{1}{1^2} \right) \\ &= -2.178 \times 10^{-18} \text{ J}(0 - 1) = 2.178 \times 10^{-18} \text{ J}\end{aligned}$$

The energy required to remove the electron from a hydrogen atom in its ground state is $2.178 \times 10^{-18} \text{ J}$.

Calculate the energy required to remove the electron from a hydrogen atom in its ground state.

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The energy required to remove the electron from a hydrogen atom in its ground state is $2.178 \times 10^{-18} \text{ J}$.

The Dual Nature of the Electron

Why the energies of the hydrogen electron are quantized?

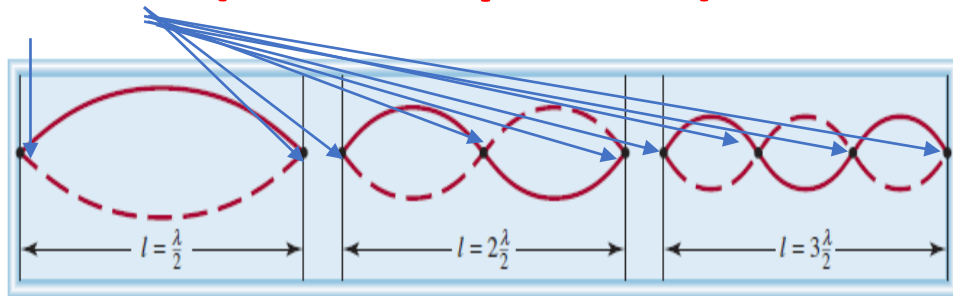
Why is the electron in a Bohr atom restricted to orbiting the nucleus at certain fixed distances?

no one, not even Bohr himself, had a logical explanation

In 1924 Louis de Broglie

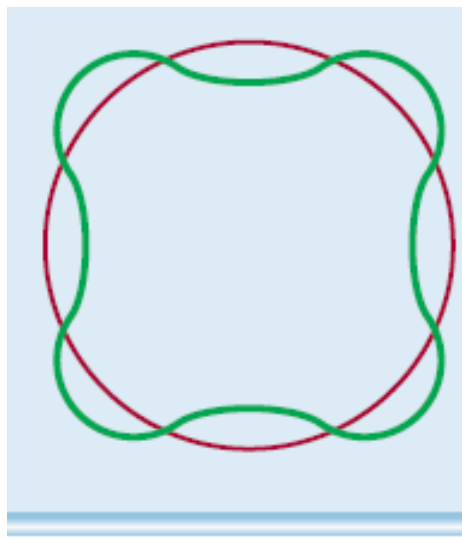
If light waves can behave like a stream of particles (photons), then perhaps particles such as electrons can possess wave properties.

Node (zero amplitude)

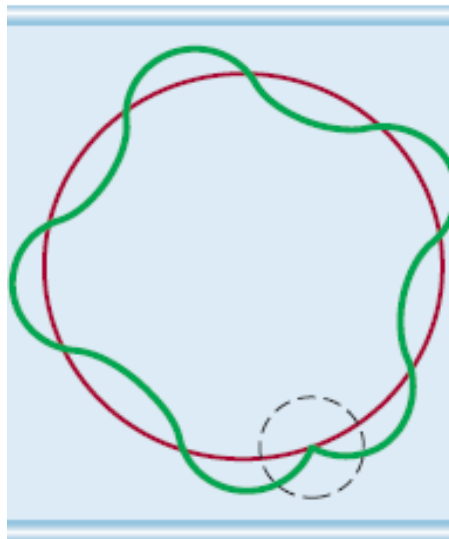


An electron bound to the nucleus behaves like a standing wave

The greater the frequency of vibration, the shorter the wavelength of the standing wave and the greater the number of nodes



(a)



(b)

if an electron does behave like a standing wave in the hydrogen atom, the length of the wave must fit the circumference of the orbit exactly.

If not

wave would partially cancel itself.
amplitude becomes zero
i.e.

Wave would not exist.

Relation between the circumference of an allowed orbit ($2\pi r$) and the wavelength (λ) of the electron is given by

$$2\pi r = n\lambda$$

r = the radius of the orbit,

λ = is the wavelength of the electron wave, and

$n = 1, 2, 3,$

De Broglie's reasoning led to the conclusion that waves can behave like particles and particles can exhibit wavelike properties.

wavelengths associated with a moving particle

$$\lambda = \frac{h}{mu}$$

mass

velocity

Calculate the wavelength of the “particle” in the following two cases: (a) The fastest serve in tennis is about 150 miles per hour, or 68 m/s. Calculate the wavelength associated with a 6.0×10^{-2} -kg tennis ball traveling at this speed. (b) Calculate the wavelength associated with an electron (9.1094×10^{-31} kg) moving at 68 m/s.

$$\begin{aligned}\lambda &= \frac{h}{mu} \\ &= \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{(6.0 \times 10^{-2} \text{ kg}) \times 68 \text{ m/s}} \\ &= 1.6 \times 10^{-34} \text{ m} \quad 1.6 \times 10^{-25} \text{ nm}\end{aligned}$$

Wavelength is very small . For this reason, the wave properties of a tennis ball cannot be detected by any existing measuring device.

$$\begin{aligned}\lambda &= \frac{h}{mu} \\ &= \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{(9.1094 \times 10^{-31} \text{ kg}) \times 68 \text{ m/s}} \\ &= 1.1 \times 10^{-5} \text{ m} \quad 1.1 \times 10^4 \text{ nm, infrared region}\end{aligned}$$

This calculation shows that only electrons (and other submicroscopic particles) have measurable wavelengths.

Calculate the energy required to excite the hydrogen electron from level $n=1$ to level $n=2$. Also calculate the wavelength of light that must be absorbed by a hydrogen atom in its ground state to reach this excited state.

An electron in the hydrogen atom makes a transition from an energy state of principal quantum numbers n_i to the $n=2$ state. If the photon emitted has a wavelength of 434 nm, what is the value of n_i ?

Calculate the maximum wavelength of light capable of removing an electron for a hydrogen atom from the energy state characterized by $n=1$, by $n=2$.

Calculate the wavelength (in nanometers) of a H atom (mass 1.674×10^{-27} kg) moving at 7.00×10^2 cm/s.

Quantum numbers

orbitals is characterized by a series of numbers called quantum numbers, which describe various properties of the orbital:

The **principal quantum number** (n) has integral values: 1, 2, 3, The principal quantum number is related to the **size and energy of the orbital**. As n increases, the orbital becomes larger and the electron spends more time farther from the nucleus. An increase in n also means higher energy, because the electron is less tightly bound to the nucleus, and the energy is less negative.

The **angular momentum quantum number** (l) has integral values from 0 to $(n - 1)$ for each value of n . This quantum number is related to the **shape of atomic orbitals**. The value of , for a particular orbital is commonly assigned a letter: , $l=0$ is called s; , $l=1$ is called p; , $l=2$ is called d; , $l=3$ is called f.

The magnetic quantum number (m_l) has integral values between l and $-l$ including zero. The value of m_l is related to the orientation of the orbital in space relative to the other orbitals in the atom.

Value of ℓ	0	1	2	3	4
Letter Used	<i>s</i>	<i>p</i>	<i>d</i>	<i>f</i>	<i>g</i>

n	ℓ	Sublevel Designation	m_ℓ	Number of Orbitals
1	0	1 <i>s</i>	0	1
2	0	2 <i>s</i>	0	1
	1	2 <i>p</i>	-1, 0, +1	3
3	0	3 <i>s</i>	0	1
	1	3 <i>p</i>	-1, 0, 1	3
	2	3 <i>d</i>	-2, -1, 0, 1, 2	5
4	0	4 <i>s</i>	0	1
	1	4 <i>p</i>	-1, 0, 1	3
	2	4 <i>d</i>	-2, -1, 0, 1, 2	5
	3	4 <i>f</i>	-3, -2, -1, 0, 1, 2, 3	7

Spin Quantum Number (m_s) : This quantum number refers to the two possible orientations of the spin axis of an electron; possible values are $+1/2$ and $-1/2$.

For principal quantum level $n = 5$, determine the number of allowed subshells (different values of l), and give the designation of each.

Pauli Exclusion Principle

Hund's Rule

Aufbau Principle

The History of the Periodic Table

The periodic table was originally constructed to represent the patterns observed in the chemical properties of the elements.

The first chemist to recognize patterns was Johann Dobereiner (1780–1849), who found several groups of three elements that have similar properties, for example chlorine, bromine, and iodine. However, as Dobereiner attempted to expand this model of triads (as he called them) to the rest of the known elements, it became clear that it was severely limited.

The next notable attempt was made by the English chemist John Newlands, who in 1864 suggested that elements should be arranged in octaves, based on the idea that certain properties seemed to repeat for every eighth element in a way similar to the musical scale, which repeats for every eighth tone

The present form of the periodic table was conceived independently by two chemists: the German Julius Lothar Meyer (1830–1895) and Dmitri Ivanovich Mendeleev (1834–1907). Usually, Mendeleev is given most of the credit, because it was, he who emphasized how useful the table could be in predicting the existence and properties of still unknown elements



TABELLE II								
REIHEN	GRUPPE I. — R ² O	GRUPPE II. — RO	GRUPPE III. — R ² O ³	GRUPPE IV. RH ⁴ RO ²	GRUPPE V. RH ³ R ² O ⁵	GRUPPE VI. RH ² RO ³	GRUPPE VII. RH R ² O ⁷	GRUPPE VIII. — RO ⁴
1	H = 1							
2	Li = 7	Be = 9,4	B = 11	C = 12	N = 14	O = 16	F = 19	
3	Na = 23	Mg = 24	Al = 27,3	Si = 28	P = 31	S = 32	Cl = 35,5	
4	K = 39	Ca = 40	= 44	Ti = 48	V = 51	Cr = 52	Mn = 55	Fe = 56, Co = 59, Ni = 59, Cu = 63.
5	(Cu = 63)	Zn = 65	= 68	= 72	As = 75	Se = 78	Br = 80	
6	Rb = 85	Sr = 87	? Yt = 88	Zr = 90	Nb = 94	Mo = 96	= 100	Ru = 104, Rh = 104, Pd = 106, Ag = 108.
7	(Ag = 108)	Cd = 112	In = 113	Sn = 118	Sb = 122	Te = 125	J = 127	
8	Cs = 133	Ba = 137	? Di = 138	? Ce = 140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	? Er = 178	? La = 180	Ta = 182	W = 184	—	Os = 195, Ir = 197, Pt = 198, Au = 199.
11	(Au = 199)	Hg = 200	Tl = 204	Pb = 207	Bi = 208	—	—	— — — —
12	—	—	—	Th = 231	—	U = 240	—	— — — —

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Table 7.3 | Comparison of the Properties of Germanium as Predicted by Mendeleev and as Actually Observed

Properties of Germanium	Predicted in 1871	Observed in 1886
Atomic weight	72	72.3
Density	5.5 g/cm ³	5.47 g/cm ³
Specific heat	0.31 J/(°C · g)	0.32 J/(°C · g)
Melting point	Very high	960°C
Oxide formula	RO ₂	GeO ₂
Oxide density	4.7 g/cm ³	4.70 g/cm ³
Chloride formula	RCl ₄	GeCl ₄
Boiling point of chloride	100°C	86°C

Figure 7.25 | Mendeleev's early periodic table, published in 1872. Note the spaces left for missing elements with atomic masses 44, 68, 72, and 100.
(From *Annalen der Chemie und Pharmacie*, VIII, Supplementary Volume for 1872, page 511.)

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For example, in 1872 when Mendeleev first published his table (Fig. 7.25), the elements gallium, scandium, and germanium were unknown. Mendeleev correctly predicted the existence and properties of these elements from gaps in his periodic table. The data for germanium (which Mendeleev called ekasilicon) are shown in Table 7.3. Note the excellent agreement between the actual values and Mendeleev's predictions, which were based on the properties of other members in the group of elements similar to germanium.

Periodic Trends in Atomic Properties

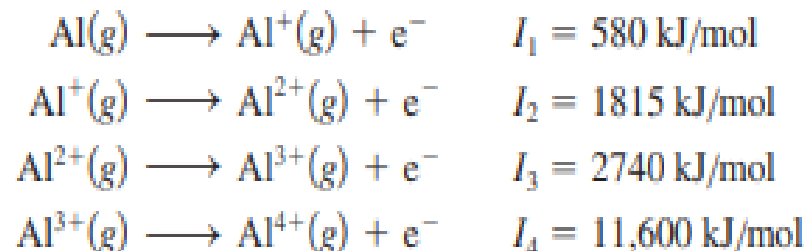
Ionization Energy

Ionization energy is the energy required to remove an electron from a gaseous atom or ion:



where the atom or ion is assumed to be in its ground state.

To introduce some of the characteristics of ionization energy, we will consider the energy required to remove several electrons in succession from aluminum in its gaseous state. The ionization energies are



The first ionization energy for phosphorus is 1060 kJ/mol, and that for sulfur is 1005 kJ/mol. Why?

Table 7.5 | Successive Ionization Energies (kJ/mol) for the Elements in Period 3

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	495	4560					
Mg	735	1445	7730	Core electrons*			
Al	580	1815	2740	11,600			
Si	780	1575	3220	4350	16,100		
P	1060	1890	2905	4950	6270	21,200	
S	1005	2260	3375	4565	6950	8490	27,000
Cl	1255	2295	3850	5160	6560	9360	11,000
Ar	1527	2665	3945	5770	7230	8780	12,000

*Note the large jump in ionization energy in going from removal of valence electrons to removal of core electrons.